INCH-POUND

MIL-C-71090 (AR)
15 November 1991

MILITARY SPECIFICATION

CANISTER SMOKE, ASSEMBLY FOR PROJECTILE, PRACTICE, 155MM, M804E1

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This military specification covers parts for assembling and packing of Canister Smoke, Assembly (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1320

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STANDARDS

MILITARY

MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1218	- ACS Chemicals

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER, (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12913950	- Canister Smoke, Assembly

(Copies of drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the Contracting activity or as directed by the Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (See 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E300	- Recommended Practice for Sampling Industrial Chemicals
ASTM D2905	- Statement on Number of Specimens Required to Determine the Average Quality of Textile

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Canister Smoke, Assembly. The assembly and components shall comply with all requirements specified on Drawing (Dwg.) 12913950 and with all requirements specified in applicable specifications.

3.3 Air pressure test of canister smoke assembly. The complete canister smoke assembly shall withstand a minimum air pressure of 5 psi for a minimum of 15 seconds without leaking as evidenced by a drop in gage pressure of 0.50 psi maximum.

3.4 Moisture content. The moisture content of the smoke mix, in the sealed container, shall be 0.1% maximum.

3.5 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, dust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. The finished mix shall be uniform and free from lumps or the tendency to form lumps.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Canister Smoke, Assembly	12913950	20
Pad	9349934	5
Cap	9349935	5
Canister	9349936	5

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.2.1 Smoke mix. Five (5) of the assemblies shall be opened and the mix in each tested for composition and moisture content.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
.	Canister Smoke, Assembly and Components	1 OF 1	
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Pad Canister Smoke, Assembly (Dwg 12913950) Examination for defects	20, 0-1	3.2	4.4.2.4
	Components (Dwg 9349934) Examination for defects	5, 0-1	3.2	4.4.2.1
	Cap (Dwg 9349935) Examination for defects	5, 0-1	3.2	4.4.2.2
	Canister (Dwg 9349936) Examination for defects	5, 0-1	3.2	4.4.2.3

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations and tests.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below,using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II Attributes Sampling Inspection

Lot size	Inspection Level					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Number under inspection levels indicate inspection sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.11). When required, the contractor shall demonstrate that the effectiveness of the proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71090 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Pad	1 OF 1	9349934
			NEXT HIGHER ASSEMBLY: 12913950

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness	0-1, L II	3.2	Gage
Minor				
201	Diameter	0-1, L IV	3.2	Gage
202	Evidence of poor workmanship	0-1, L IV	3.6	Visual

NOTES:

AMSMC Form 1570a, 1 Jul 89 — Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cap	1 OF 1	9349935
			NEXT HIGHER ASSEMBLY: 12913950

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Small diameter of shoulder	0-1,L II	3.2	Gage
102	Thickness	0-1,L II	3.2	Gage
Minor				
201	Outside diameter	0-1,L IV	3.2	Gage
202	Depth of shoulder	0-1,L IV	3.2	Gage
203	Evidence of poor workmanship	0-1,L IV	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71090 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Canister	1 OF 1	9349936
			NEXT HIGHER ASSEMBLY
			12913950

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter at mouth	0-1,L II	3.2	Gage
102	Outside diameter at crimp flange	0-1,L II	3.2	Gage
103	Thickness through wall, min.	0-1,L II	3.2	Gage
104	Length straight at neck, min.	0-1,L II	3.2	Gage
Minor				
201	Outside diameter of body, min.	0-1,L IV	3.2	Gage
202	Thickness through base, min.	0-1,L IV	3.2	Gage
203	Length total	0-1,L IV	3.2	Gage
204	Evidence of poor workmanship	0-1,L IV	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71090 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Canister Smoke, Assembly	1 OF 1	12913950
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Air leak test	Ø-1,L II	3.3	4.5.3
102	Mix composition incorrect	Note 1	3.2	4.5.1.1
103	Moisture content over max.	Note 2	3.4	4.5.2
104	Weight of charge in body	Ø-1,L II	3.2	Balance
105	Outside dia. of container, max.	Ø-1,L II	3.2	Gage
106	Outside dia. of crimp, max.	Ø-1,L II	3.2	Gage
107	Crimp thickness	Ø-1,L II	3.2	Gage
108	Any component missing or incorrectly assembled	Ø-1,L II	3.2	Visual
Minor				
201	Crimp height	Ø-1,L IV	3.2	Gage
202	Depth from top of crimp to cap well	Ø-1,L IV	3.2	Gage
203	Double crimp not 360 degrees	Ø-1,L IV	3.2	Visual
204	Length, total to end of crimp	Ø-1,L IV	3.2	Gage
205	Length, total to end of pad, max.	Ø-1,L IV	3.2	Gage
206	Sealing compound excessive, inadequate or missing	Ø-1,L IV	3.2	Gage
207	Marking missing, misleading or unidentifiable	Ø-1,L IV	3.2	Visual
208	Evidence of poor workmanship	Ø-1,L IV	3.6	Visual

NOTES:
1. Sampling in accordance with 4.4.3.1.
2. Sampling in accordance with 4.4.3.2.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Packing	1 OF 1	9344016
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Any spacer missing, prior to sealing box	Ø-1,L IV	3.2	Visual
202	Incorrect number of units in box, prior to sealing	Ø-1,L IV	3.2	Visual
203	Box improperly sealed	Ø-1,L IV	3.2	Visual
204	Box improperly marked	Ø-1,L IV	3.2	Visual
205	Evidence of poor workmanship	Ø-1,L IV	3.6	Visual

NOTES:

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4.4.3. Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

PRECAUTION WARNING

This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.3.1 Compositions.

4.4.3.1.1 Batch inspection. Each batch (see 6.8) of smoke mix shall be inspected for composition, if process inspection is not opted. Acceptance for this test shall be in accordance with MIL-STD-414, Table B-3, Inspection Level I, AQL 1.0%, with the inspection lot equal to each batch of mix. Lot size shall be the number of containers, ± 10%, to be filled from the batch. The samples shall be drawn from filled bodies. Total percent defective shall be calculated for each constituent. If any of the constituents fail to meet the acceptability criterion, the batch shall be rejected.

4.4.3.1.2 Process inspection.

4.4.3.1.2.1 Process qualification. The initial batches (see 6.8) of smoke mix from each blender to be used in production shall be tested for composition. Acceptance for this test shall be in accordance with MIL-STD-414, Table B-3, Inspection Level I, AQL 1.0%, with the inspection lot equal to each batch of mix. Lot size shall be the number of containers, ± 10%, to be filled from the batch. The samples shall be drawn from filled bodies. Total percent defective shall be calculated for each constituent. If any of the constituents fail to meet the acceptability criterion, the batch shall be rejected. The process shall be considered qualified when five consecutive batches have been accepted from each blender to be used in production.

4.4.3.1.2.2 Process control. At the option of the contractor, process control may be substituted for batch inspection after qualification per 4.4.3.1.2.1. Process control shall be as follows:

a. During process qualification, record the blender speed and blending time for each batch.

b. Prepare control limits for blending speed and time based on recorded data from the five (5) qualifying batches (see 6.10).

c. Weigh and check weigh each constituent.

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d. Control blending speed and time within the established limits.

If blending speed or time is outside the control limits, sample inspection for composition of that batch shall be conducted per 4.4.3.1.1. If blending speed or time must be adjusted to fall outside the control limits, re-qualification shall be conducted per 4.4.3.1.2.1. Re-qualification shall also be required, if in the opinion of the Government, maintenance or repair to the blender or dispensing mechanism effects process control parameters.

4.4.3.1.2.3 Verification. When process control is in effect, a verification inspection shall be conducted on one batch from each seven (7) batches from each blender. Sample batch inspection shall be per 4.4.3.1.1. If a sample batch is rejected, re-qualification of the process per 4.4.3.1.2.1 shall be required before process control is resumed.

4.4.3.2 Moisture content. Each shift's production shall be tested for moisture content. Acceptance for this test shall be in accordance with MIL-STD-414, Table B-3, Inspection Level I, AQL 1.0%, with the inspection lot equal to each shift's loading. Lot size shall be the number of containers, ± 10%, to be filled and sealed during one production shift. The samples shall be drawn from filled containers prior to sealing. If percent defective calculated fails to meet the acceptability criterion, the shift's production shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform, the examinations and tests prescribed herein is described in the Inspection Method Reference column in the table starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Analytical test methods and procedures (see 6.9). The following tests shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical text books. ASTM Method D2905 may also be used to determine the number of replicates required. Unless otherwise specified herein, all chemicals and reagents used in test methods shall be ACS grade in accordance with MIL-STD-1218 or reagent grade.

4.5.1.1 Composition.

4.5.1.1.1 Potassium nitrate. Into a tared 150 milliliter (ml) beaker accurately weigh to the nearest 0.0001 gram (g) approximately

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3g of smoke composition. Add 40 ml of distilled water and continuously swirl the contents of the beaker for 5 minutes. Allow the solid material to settle to the bottom of the beaker. Fit a Fisher Filtrator or equal with a clean glass medium porosity filtering crucible and a 250 ml beaker to receive the filtrate. Adjust the filtrator vacuum so the filtrate does not splash out of the receiver flask and decant the supernatant liquid from the 150 ml beaker through the crucible and into the 250 ml receiver beaker. Wash the solid to the bottom of the flask with a stream of distilled water (10 ml or less) and again swirl the contents of the 150 ml beaker for two minutes. Allow the residue to settle to the bottom of the beaker and decant the liquid through the filtering crucible into the 250 ml beaker. Wash the residue using this procedure two more times, transferring the solid residue to the filtering crucible on the final wash (quantitative transfer not necessary). Rinse the crucible and crucible holder with a small amount of distilled water catching the rinse water in the 250 ml beaker. Transfer the filtrate and washings to a 100 ml volumetric flask, dilute to the mark with distilled water and mix thoroughly. Fill a 1.0 centimeter (cm) path length silica spectra photometric cell with the solution from the volumetric flask and fill the matching cell with distilled water. Using the distilled water as the blank, measure the absorbance of the solution at 301 nanometer (nm) using a Beckman Model DU Spectrophotometer or equal.

Prepare standards for the potassium nitrate determination as follows: Weigh the approximate amounts given in Table III, to the nearest 0.0001 g in a clear, dry, tared 150 ml beaker.

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TABLE III

Potassium Nitrate Standardization Solutions

Standard No.	Potassium Nitrate, g	Potassium perchlorate, g
1	0.570 - 0.585	0.600
2	0.600	0.600
3	0.615 - 0.630	0.600

Add 40 ml of distilled water to each standard and follow the same procedure given for the analysis of the sample.

Calculate the percent potassium nitrate using the following equations:

Standardization Values:

$$b = \frac{(\Sigma A)\ (\Sigma c^2) - (\Sigma Ac)\ (\Sigma c)}{N\ (\Sigma c^2) - (\Sigma c)^2}$$

and

$$m = \frac{(\Sigma A - Nb)}{(\Sigma c)}$$

where A = Absorbance of the standard solutions.
b = The A intercept.
c = The concentration of potassium nitrate in g/100 ml.
m = The slope of the absorbance vs. concentration data.
and N = The number of standard samples tested.

Percent Potassium Nitrate:

$$\text{Potassium nitrate, \%} = \frac{100\left(\frac{A-b}{m}\right)}{W - WM}$$

where A = The absorbance obtained for the sample.
b & m = The values obtained from the standardization calculation above.
W = The sample weight in g.
and M = The percent moisture obtained in 4.5.2 expressed as a decimal.

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4.5.1.1.2 Determination of potassium perchlorate. Into a tared 30 ml medium porosity glass filtering crucible weight to the nearest 0.0001g an approximately 1.3g sample of smoke composition. Place a clean, dry glass stirring rod in the crucible and weigh the crucible, sample and stirring rod to the nearest 0.0001g. Place the crucible and contents in a crucible holder attached to a suction flask and add a 15 ml portion of distilled water. Allow the water to remain in contact with the sample for 5 minutes, stirring immediately after the water addition, after 2-3 minutes and just prior to turning on the vacuum to draw the water through the crucible. Use the glass stirring rod for the stirring operations, being careful to avoid grinding the metal particles against the sides or bottom of the crucible. Repeat the extraction procedure four more times with 15 ml portions of distilled water allowing the water to remain in contact with the sample for three minutes in each case. Stir the mixture immediately after water addition, after 1.5 minutes and prior to applying vacuum. Rinse the crucible side and contents with a stream of distilled water and aspirate. Wash the contents of the crucible with about 5 ml of 95% ethanal and aspirate. Wash the crucible contents with about 5 ml of ether and aspirate until the ether odor is no longer noticeable. Dry the crucible stirring rod, and contents, for a half hour at 100 degrees Celsius (C) + 2 degrees C. Cool in a desiccator and weigh to the nearest 0.0001g. Reserve the crucible stirring rod and residue for the determination of zinc and aluminum. Calculate the percent potassium perchlorate as follows:

$$\% \text{ Potassium Perchlorate} = \frac{(A - B)\ 100}{W - WM} - C$$

where

A = The weight of the sample, crucible and glass rod prior to water extraction in g.

B = The weight of the residue, crucible and glass rod after water extraction in g.

W = The weight of the sample in g.

M = The percent moisture as determined in 4.5.2 expressed as a decimal.

C = The percent potassium nitrate as determined in 4.5.1.1.1

4.5.1.1.3 Determination of zinc and aluminum. Place the crucible containing the zinc and aluminum residue reserved from the potassium perchlorate determination in a crucible holder attached to a suction flask. Add 20 ml of 5% nitric acid (5 ml concentrated nitric acid added to 95 ml distilled water), which has been cooled to 2 degrees Celsius + 1 degrees Celsius, to the crucible and stir.

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Allow the acid to remain in contact with the residue for five minutes, stirring after 2-3 minutes. Apply vacuum to remove the acid from the crucible. Add a second 20 ml portion of cold (2 degrees Celsius + 1 degrees Celsius) 5% nitric acid and repeat the extraction procedure. Perform four additional extractions using room temperature 10% nitric acid (10ml nitric acid and 90 ml distilled water) stirring and aspirating each in the same manner as was done with the 5% nitric acid extractions. Wash the residue with several small portions of distilled water to remove any excess acid and aspirate. Discard the acid washings from the suction flask. Replace the crucible in the suction flask and wash the residue with 5 ml of 95% ethanol and aspirate. Wash the residue with 5 ml of ether and aspirate until the odor of ether is not longer noticeable. Dry the crucible contents and stirring rod in an oven at 100 degrees Celsius $\pm$ 2 degrees Celsius for a half hour, cool in a deseccator and weigh to the nearest 0.0001g.

Calculate the percent zinc and aluminum as follows:

$$\% \text{ zinc} = \frac{100\ (B - D)}{W - WM}$$

and

$$\% \text{ aluminum} = \frac{100\ (D - E)}{W - WM}$$

where B = The weight of the residue, crucible and glass rod after water extraction in g (see 4.5.1.1.2)

D = The weight of the crucible residue and glass rod after the nitric acid extraction in g.

W = The weight of the sample in g.

M = The percent moisture as determined in 4.5.2.

E = The tare weight of the crucible plus the stirring rod in g.

4.5.2 Moisture content. Transfer an accurately weighed portion of approximately 5g of the sample to a tared weighing bottle. Dry the bottle and contents at 100 degrees Celsius + 5 degrees Celsius for 2 hours minimum, cool in a desiccator and weigh. Calculate the loss in weight as percent moisture.

4.5.3 Air pressure test canister smoke assembly. The complete assembly shall be subjected to an air pressure test using test equipment in accordance with 4.4.4. Observation shall be made for compliance with requirements of 3.3.

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5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The smoke canister covered by this specification is intended for use in spotting of artillery practice projectiles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirements.

d. Packaging requirements, if other than specified in section 5.

e. Serialization requirements, if applicable.

f. Certification of conformance for each lot or shipment of product.

g. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.1.2).

6.3 Submission of contractor inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

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6.5 Visual examination qualification. When compliance with the acceptable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.6 Submission of test data and ammunition data cards. In addition to the normal distribution of records, when the projectile is procured by the US Army Armament, Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: PMCAR-CCH-P and SMCAR-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000 and Commander, ARRCOM, ATTN: AMSMC-QAD, Rock Island, IL 61299-6000.

6.7 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Command (ARRADCOM) or US Army Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.8 Batch. A batch is defined as a quantity of ingredients blended in a single container at one time.

6.9 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARDEC, ATTN: SMCAR-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.10 Computing process control limits. The control limits shall be: $\overline{X} \pm 1.342\sigma$.

where: $\overline{X}$ = average blending time or speed for 5 consecutive acceptable batches.

σ = standard deviation of 5 times or speeds calculated as $\sigma = \sqrt{\frac{\Sigma X^2 - (\Sigma X)^2}{5}}$

with X being the individual batch blending time or speed.

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6.11 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.12 Subject term (keyword) listing.

Air pressure test
Ammunition
Moisture content of smoke mix

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-AC68)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-71090 (AR)	15 NOVEMBER 1991

3. DOCUMENT TITLE
CANISTER SMOKE, ASSEMBLY FOR PROJECTILE, PRACTICE, 155MM, M804E1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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